

Project Title:

*Exploration of Neural Progenitor Cell Heterogeneity and Fate Decisions Using
Single-Cell Transcriptomics*

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Abstract

The mammalian cerebellum develops from a group of distinct cell types that undergo rapid proliferation and differentiation during embryogenesis. 34,230 cerebellar mouse cells identified using single-cell RNA-sequencing from twelve developmental timepoints were analyzed to replicate and extend current knowledge of cerebellar development. Specialized cell types and progenitor populations were identified using known genetic marker genes and their trajectory was inferred using pseudotemporal ordering. Analysis of the single cell data was performed in RStudio using Seurat and Monocle3. Distinct populations of early progenitor cells were identified, and key marker genes and gene co-expression dynamics were uncovered. Genes involved in the differentiation of neural progenitor cells to GABAergic progenitors and non-neuronal glial cells were identified. This work provides new insight into the genes and molecular mechanisms involved in fate progression of early progenitors into lineage specific progenitor types and those involved in more mature fate decisions that give rise to non-neuronal lineages. A comprehensive, reproducible and open-source workflow was developed to enable customizable analysis of novel developmental single-cell RNA-sequencing datasets.

1.0 Introduction

The neural tube is the embryonic precursor to the central nervous system (CNS), which gives rise to the cerebellum through the metencephalon. The cerebellum is the main motor coordination centre of the CNS and recent studies have linked it to several cognitive functions, including executive function and spatial cognition (Wang & Zoghbi, 2001). Although the cerebellum contains about 10% of the total brain volume, a fully developed cerebellum contains over 50% of the total neurons found in the CNS. Cerebellar development is a prolonged process initiating early in embryogenesis and continuing after birth (Wang & Zoghbi, 2001). This

extensive developmental window makes the cerebellum vulnerable to neurodevelopmental abnormalities, including Chiari and Dandy-Walker malformations, as well as cellular differentiation related cerebellar neoplasias (Wang & Zoghbi, 2001).

Neural progenitor cells (NPCs) are self-renewing, multipotent stem cells that differentiate into neurons, glial cells and oligodendrocytes during nervous system development (Carter et al., 2018). During embryogenesis, NPCs undergo rapid proliferation and differentiation, supporting the growth and development of organs, in part, by giving rise to distinct cell subtypes (Li & Kirschner, 2014). The increase in size and diversity of the cerebellum is driven by two main germinal zones, the rhombic lip (RL) and ventricular zone (VZ), which give rise to *Atoh1* (Atonal homologue 1) and *Ptfla* (Pancreas associated transcription factor) NPCs, respectively (Carter et al., 2018). Progenitors from the RL give rise to excitatory glutamatergic neurons, whereas those from the VZ produce inhibitory GABAergic neurons. Previous work has shown that in *Ptfla* knockout mice, VZ progenitors migrate directly to the external granule layer (EGL) and take on granule cell phenotypes. This demonstrates that *Ptfla* expression is required to specify a GABAergic identity (Pascual et al., 2007).

The emergence of distinct neuronal lineages at specific time periods has been established in previous mouse models. For example, GABAergic progenitors expressing *Ptfla* have been detected in the cerebellar primordium between embryonic days E12.5 and E14.5 (Hoshino et al., 2005). *Atoh1* expression has been observed in rhombic lip progenitors between E9 and E11, and again at E12 as the RL begins to produce mainly granule neuron progenitors (GNPs), which undergo rapid proliferation, peaking at postnatal day P5 (Machold et al., 2011; Carter et al., 2018). While this understanding is important, it does not reveal how molecular dynamics change over time or whether transitional cell populations arise during fate specification.

Single-cell RNA-sequencing (scRNA-seq) is an effective genomic technique for the identification and quantification of messenger RNA (mRNA) transcripts at the individual cell level (Haque et al., 2017). ScRNA-seq enables the characterization of cell type heterogeneity within a given tissue sample and in recent years, has successfully been used to profile cell subtypes in the developing mouse cerebellum. While the field has advanced significantly, the transcriptional mechanisms driving early NPC commitment to GABAergic or glutamatergic progenitor identities, and their divergence towards non-neuronal lineages, remain poorly defined.

The analysis in this project was guided by two hypotheses: (1) Cerebellar NPCs arising at E10.5-E11.5 remain plastic for stochastic fate switching between *Atoh1* RL progenitors and *Ptfla* VZ progenitors; and (2) Bergmann glia (BG) initially overlap with GABAergic progenitors in the VZ between E13-E14 but activate gliogenesis genes to establish a distinct non-neuronal trajectory. Based on this, the aims of this project were to identify and characterize distinct early NPC populations in embryonic mouse cerebellum scRNA-seq data and to determine key genes involved in the differentiation of NPCs into lineage specific progenitor types and non-neuronal cell types. Distinct marker genes and functional difference between early NPC populations were identified using Gene Ontology (GO) analysis. Pseudotemporal analysis was performed to track changes in key genes across pseudotime and to visualize how gene expression dynamics differ between GABAergic and glial lineages.

Many aspects of the molecular mechanisms guiding cell differentiation remain to be discovered. Through the identification of robust subclusters of early NPCs, and the dynamics involved in lineage bifurcation points in NPCs, this work will expand on findings made by Carter et al. (2018). The proposed work will yield a more complete and detailed transcriptional map of the developing mouse cerebellum and highlight the power of pseudotime analysis when working

with scRNA-seq data. Through this, we gain valuable insight into early-stage molecular mechanisms driving cerebellar lineage specification and trajectories, providing the opportunity to better understand neurodevelopmental disorders that arise in the early stages of embryogenesis. The workflow used for this analysis combines a standard Seurat scRNA-seq workflow with Monocle3 pseudotime and trajectory analysis to present a comprehensive open-source pipeline that can be applied to any developmental scRNA-seq dataset.

2.0 Methods

2.1 GitHub Repository

The comprehensive RMarkdown analysis script, raw data files, and tables/objects produced at key steps have been uploaded to the following public GitHub Repository:

<https://github.com/vladbularca/Murine-Cerebellum-scRNA-Analysis>.

2.2 Computational Tools

This analysis was completed in RStudio, using Seurat package (v5.3.0) for normalization, dimensionality reduction, and clustering of single-cell RNA-seq data. Loupe Browser (v9.0, 10x Genomics) was used for data exploration and visualization of gene expression patterns. For pseudotime and trajectory inference, Monocle3 (v1.4.26) was employed, following the reversed graph embedding framework.

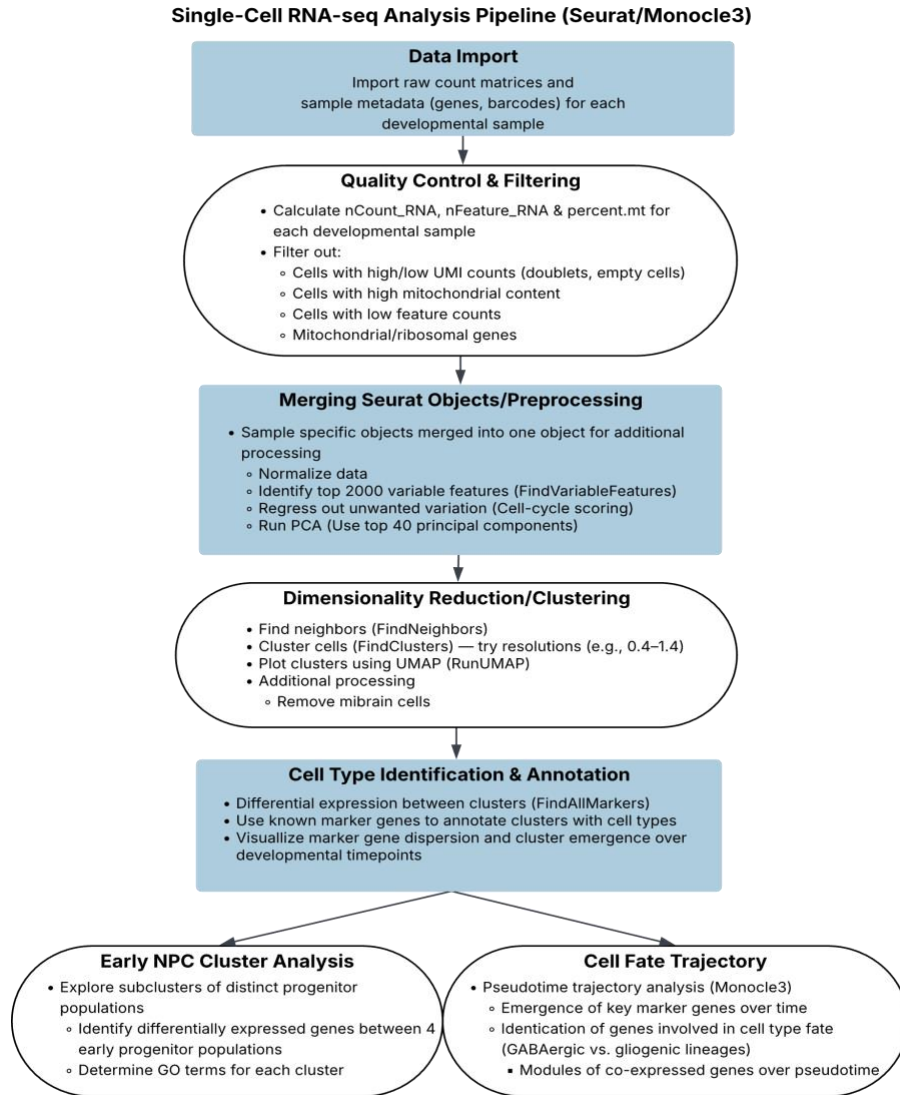


Figure 1. Analytical Pipeline Flowchart

Simplified analysis workflow highlighting key filtering, processing and hypothesis testing steps. Workflow was developed in RStudio and was adapted from Seurat & Monocle3 tutorials. A detailed breakdown of each section can be found in the RMarkdown file on GitHub.

2.3 Data Retrieval

The scRNA-seq data used for this analysis was produced by Carter et al. (2018) and is derived from tissue samples used in the paper: *A Single-Cell Transcriptional Atlas of the Developing Murine Cerebellum*. Tissue was sampled at twelve developmental timepoints of cerebellar development during early embryogenesis, ranging from E10 to P10. Two biological

mouse replicates were used from independent pregnancies, resulting in 24 total samples. The libraries were prepared using the Chromium v1 Single Cell 30 Library and Gel Bead Kit (10x Genomics) (Carter et al., 2018). The associated data was retrieved from GEO (accession GSE224466), and each sample contained corresponding barcodes.tsv, genes.tsv and matrix.mtx files. The barcodes.tsv file contains the complete list of barcodes, which act as unique identifiers for individual cells and correspond to the columns in the matrix.mtx file. The same set of barcodes were used when sequencing each developmental timepoint. The genes.tsv file contains Ensembl IDs and their corresponding gene symbols. Lastly, the matrix.mtx file is a Matrix Market coordinate file, which contains unique molecular identifier (UMI) counts per gene per cell. The 3 files for each sample were produced using Cell Ranger (10x Genomics), a package used to perform sample demultiplexing, barcode processing, and gene transcript quantification from raw single cell data.

2.4 Data Processing

The main data folder containing the necessary files for all 24 samples was imported into RStudio and a Seurat object was created for each sample. The result was a Seurat list containing 24 Seurat objects, representative of each of the 24 samples. Several data filtering steps were applied to ensure high-quality data for downstream processing. First, the mitochondrial genes were identified using the prefix “mt”, and a “percent.mt” column was added to each Seurat object. Additionally, ribosomal genes identified using the prefix “Rps|Rpl” and non-coding duplicates and pseudogenes were identified using prefixes “ka|kc|-ps”. Only cells with a mitochondrial gene percentage less than 10% were kept, as high mitochondrial percentage is an indicator of cell damage/death. After this step, all ribosomal and mitochondrial genes were removed from the cells as their presence has been shown to impact downstream clustering results

(Carter et al., 2018). `NCount_RNA` represents the total number of Unique Molecular Identifiers (UMIs) detected in a cell, after duplicate RNA molecules have been removed through UMI counting. Cells with a `nCount_RNA` less than 3,200 and greater than 15,000 were filtered out to account for empty and duplicate cells. Lastly, cells with a `nFeature_RNA` (number of genes detected in a cell) count of less than 500 were filtered out and genes with 0 counts across all cells within a sample were removed before further downstream processing. Low feature counts may be indicative of technical limitations or low cell quality. Filtering choices were based on cutoffs determined by Carter et al. (2018), with minor adjustments to account for differences between the published results and the results observed in the raw data provided.

The data for each developmental sample was then merged into one comprehensive Seurat object for clustering and downstream analysis. Using built-in Seurat commands, the data was normalized, and the 2000 most variable features across the dataset were identified. To remove the confounding influence of cell cycle heterogeneity on other biological processes of interest, a cell cycle phase difference score was calculated for each cell and then regressed out when scaling the data. PCA was then run on the data with 40 principal components and the variable features identified in the previous step. Next, `FindNeighbors` was run to construct a Shared Nearest Neighbor (SNN) graph before the clustering step. The final clustering algorithm was run using a resolution of 1.4, after multiple resolution values were tested. The function `runUMAP` was used to perform Uniform Manifold Approximation and Projection (UMAP) on the dataset. This step is key to visualizing the clusters of cells in two dimensions while maintaining global and local relationships between cells. Lastly, `FindAllMarkers` was run to identify marker genes, which are genes that are differentially expressed between a given cluster and all other clusters. Marker genes were then used to annotate the corresponding cell type for each cluster. The Seurat object

which now contained UMAP embeddings was exported as a Loupe file for further visualization in Loupe Browser.

2.5 Cluster Annotation

Clusters were annotated with cell types based on key marker genes. Specific genes have been identified in previous research to be uniquely expressed and/or upregulated by specific cell types. Combinations of multiple genes were used to complete the annotation step as relying on multiple gene markers reduces ambiguity in cell type identification. The following marker genes were used to annotate specific cell types: *Slc1a3* to identify astrocytes, *Meis2* & *Lhx2* to mark glutamatergic CNs, *Ptf1a* for GABAergic Progenitors, *Lhx5* and *Blklhe23* to identify GABAergic neuron precursors, *Pax2* & *Lbx2* for GABAergic Interneurons, *Msx1* to mark roof plate cells, *Sox10* for oligodendrocytes, *Atoh1* for glutamatergic progenitors, *Pax6* & *Zic1* for GNPs and granule neurons, *Foxp2* for Purkinje progenitors and Purkinje cells, *Hist1h2ae* & *Hist1h1b* to identify radial glial cells, *Hba-a2* & *Alas2* to mark erythrocytes, *Tmem119* for microglia and *Hopx* & *Gdf10* to identify Bergman Glia (Carter et al., 2018; Khouri-Farah et al., 2022).

2.7 Midbrain Cell Removal

After clustering and annotation was completed the first time, an ambiguous cluster was identified that did not resemble a cell type of cerebellar origin. After further investigation, this was determined to be a cluster of midbrain cells. Using Loupe Browser, several midbrain marker genes including: *Tlx3*, *Isl1*, *Dlk1*, *Otx2*, *Sncg*, *Tal2*, *Phox2a*, *Phox2b* were showing co-expression in this cluster. To remove these cells, a module score column was added to the Seurat object to score each cell based on co-expression of these genes. The cells falling within the 93rd percentile based on this ‘midbrain score’ were removed from the dataset. After these cells were removed, the Seurat workflow, including PCA, clustering, and `FindAllMarkers` was redone using the

new subset of cells. The new clusters were annotated with cell types once again. At this point, the Seurat object reflected the final structure which was used for data exploration and further downstream analysis.

2.8 Early NPC Cluster Characterization

After the initial clustering step, the group of cells that were annotated as ‘Early Progenitors’, which appeared mainly during E10-E11, were subsetted for a more focused analysis. The Seurat workflow was rerun from the beginning to find distinct clusters of early progenitor cells within this group of cells. The corresponding marker gene lists for each cluster were analyzed using GO Analysis. Top marker genes for each cluster were selected using adjusted p-value and average log2 fold change, and only markers with adjusted p-values less than 0.05 were considered. To determine the top GO terms, those with the highest fold enrichment value and gene count per GO term were selected. Fold enrichment is determined by how much specific terms are over-represented in a list compared to a background set of genes.

2.9 Pseudotime Ordering and Trajectory Inference

The Seurat object data structure was converted to a CellDataSet (CDS) object using Monocle3. The UMAP and PCA embeddings were copied from the Seurat object to maintain the clustering structure. A CDS structure contains similar features to a Seurat object but is in a format that is required for pseudotime analysis. Monocle orders cells by inferring the most probable trajectory of biological progression, introducing a temporal variable known as pseudotime (Ciao et al., 2019). The `learn_graph` command was used to determine the trajectory along the UMAP, and cells were coloured based on pseudotime using `order_cells`. To gain insight into expression dynamics at key branching points and within specific subsets of the data, two main analyses were performed. First, the clusters of cells that corresponded to

GABAergic progenitors and Bergmann Glia were isolated from the larger dataset to identify key genes that show significant changes in expression between the two lineages. The GABAergic cells and glial cells were treated as two distinct clusters. The linear additive model: “~pseudotime * seurat_clusters”, was fitted to the subsetted data. The output of this model was a list of genes whose expression across pseudotime differed significantly between the two groups of cells. The top 5 genes were selected based on the ‘estimate’ column, which represents strength and direction of the expression trend over pseudotime, and the q-value (< 0.05). Gene expression over pseudotime was plotted using `plot_genes_in_pseudotime`. Lastly, `graph_test` and `find_gene_modules`, which are commands in Monocle3 used to identify modules of co-expressed genes, were run on the data representing the GABAergic and glial cells. This same analysis was done on the early NPC clusters to further understand expression dynamics in the subset of the data that emerged during early development. `Graph_test` uses Moran’s I statistic on the k-nearest-neighbor graph to find genes that show spatial correlation along the pseudotime trajectory and `find_gene_modules` groups genes with similar expression patterns together (Ciao et al., 2019).

3.0 Results

3.1 Data Exploration

To explore features of the data, summary metrics and visualizations were generated to understand the underlying structure. After reading in the data, there were initially 17,694,720 total cells across the 24 samples. Each sample run used 737,280 droplets, each containing a unique molecular barcode. The number of cells for each sample that had a transcript count of 0 ranged from 440,000 to 650,000. After filtering and midbrain cell removal, the number of cells

that remained was drastically lower than the starting count, with a total of 36,230 cells (Table 1). E12 samples had a very low cell count after filtering compared to other samples. Quality checks for each individual sample are crucial for further downstream processing as all subsequent steps are impacted by the quality of the data. nCount_RNA, nFeature_RNA, percent.mt violin plots were generated for all 24 samples to validate that quality filtering was successful (Fig. 2, 3, 4, 5). To visualize the trend between nCount_RNA and nFeature_RNA, a scatter plot of the two was generated (Fig. 5). Pearson's correlation coefficient is 0.9 for the resultant trend, which indicates a strong positive linear correlation between the two variables.

Table 1: *Number of Cells Retained per Sample After Quality Filtering*

Summary table with total number of cells passing quality filtering steps. The number of cells varies between different timepoints and replicates.

Sample	Number of Cells
PAN_CB_E10C	1,016
PAN_CB_E10D	872
PAN_CB_E11A	798
PAN_CB_E11B	2,490
PAN_CB_E12A	3
PAN_CB_E12B	52
PAN_CB_E13A	2,631
PAN_CB_E13B	3,272
PAN_CB_E14A	187
PAN_CB_E14B	412
PAN_CB_E15A	2,141
PAN_CB_E15B	1,383
PAN_CB_E16C	1,546
PAN_CB_E16D	1,869
PAN_CB_E17A	2,263
PAN_CB_E17B	1,519
PAN_CB_P0A	2,115
PAN_CB_P0B	1,363
PAN_CB_P4A	2,383
PAN_CB_P4B	2,585
PAN_CB_P7A	7,37
PAN_CB_P7B	972
PAN_CB_P10A	1,036
PAN_CB_P10B	1,001

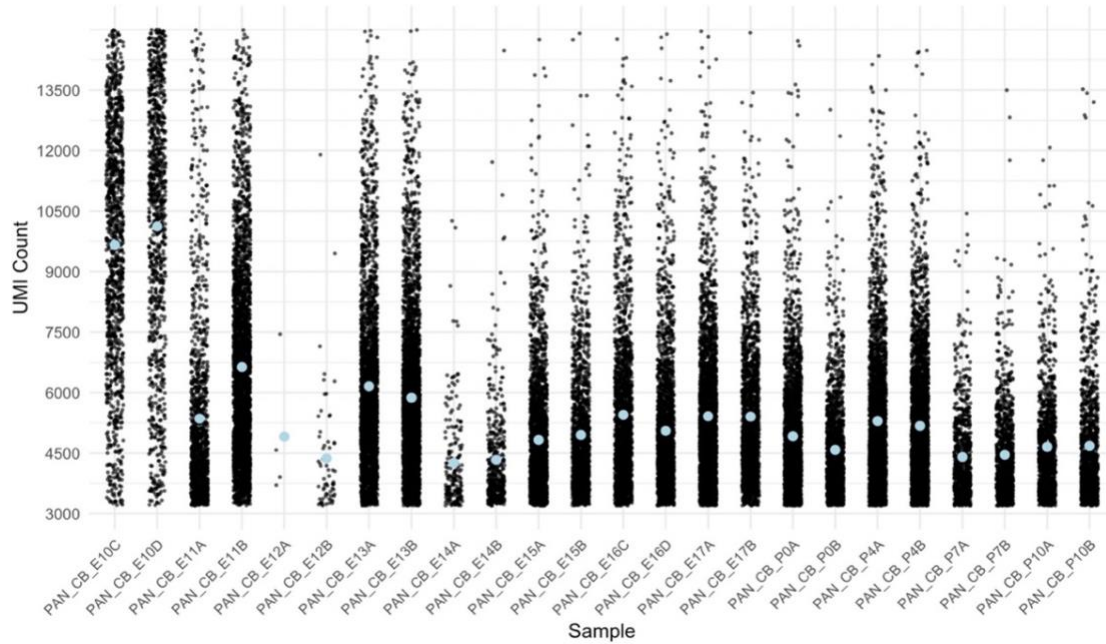


Figure 2: *Jitter Plot of UMI Count Per Sample*

Jitter plot showing total UMI counts per sample, highlighting the UMI count distribution and the abundance of cells per sample. The absence of cells with UMI counts above 15,000 and below 3000 is an indicator of successful quality filtering. The light blue points represent the mean UMI counts per sample.

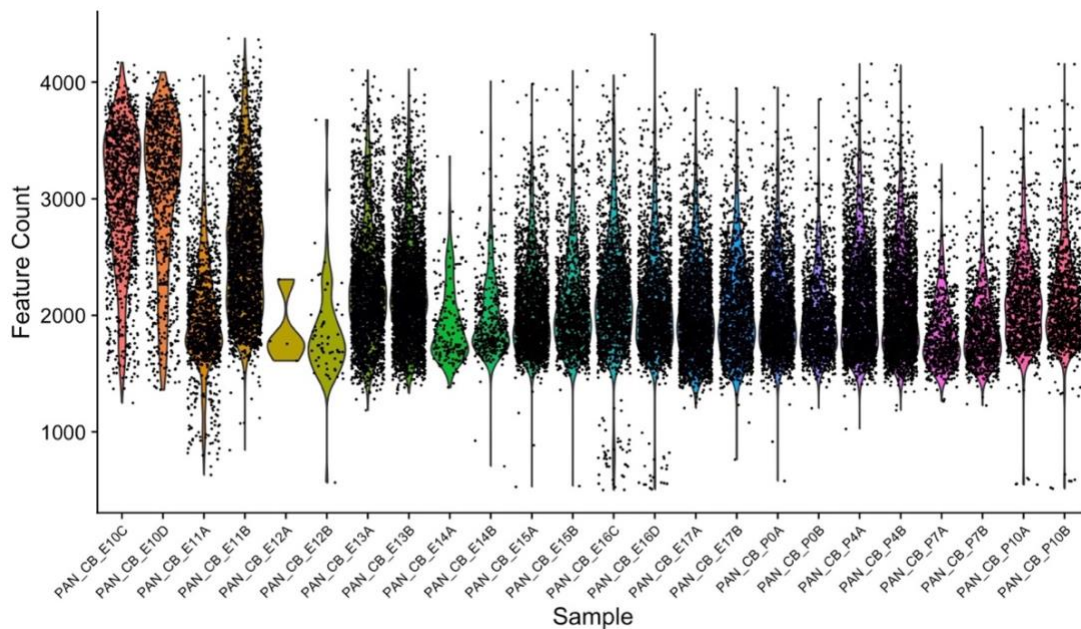


Figure 3: *Feature Count Violin Plots Per Sample*

Violin plots showing total feature counts per sample, highlighting the feature count distribution and the abundance of cells per sample. The absence of cells with feature counts below 500 is an indicator of successful quality filtering.

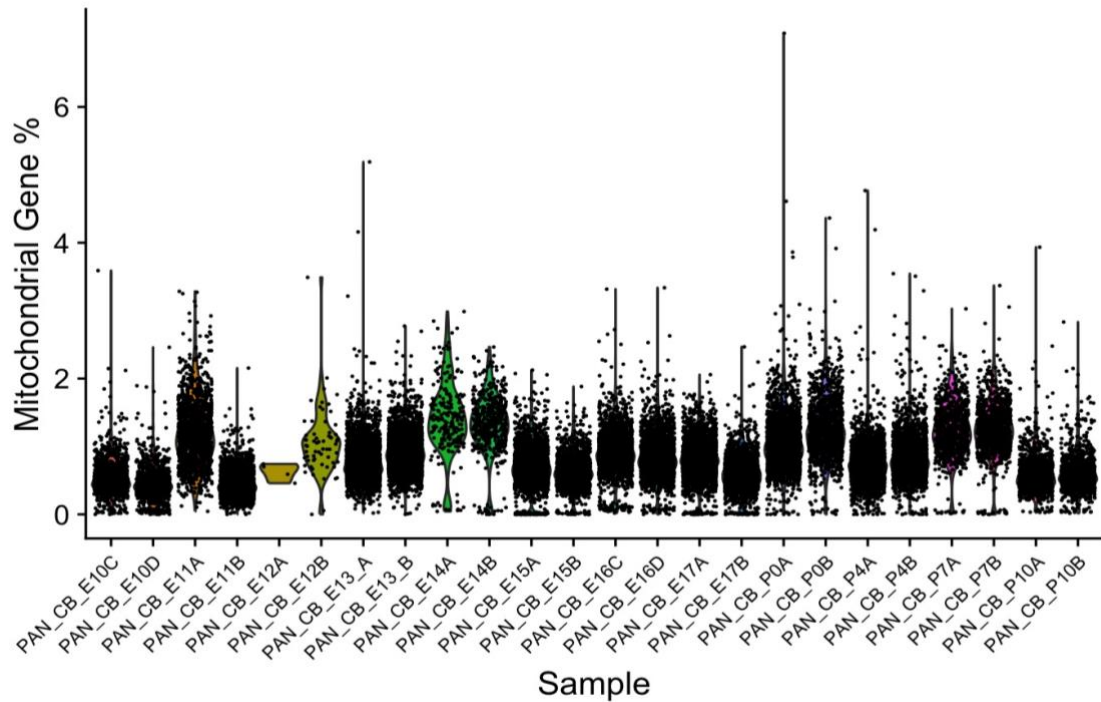


Figure 4: *Mitochondrial Gene % Violin Plots Per Sample*

Violin plots showing mitochondrial gene percentage per sample. The absence of cells with a mitochondrial gene percentage above 10% is an indicator of successful quality filtering.

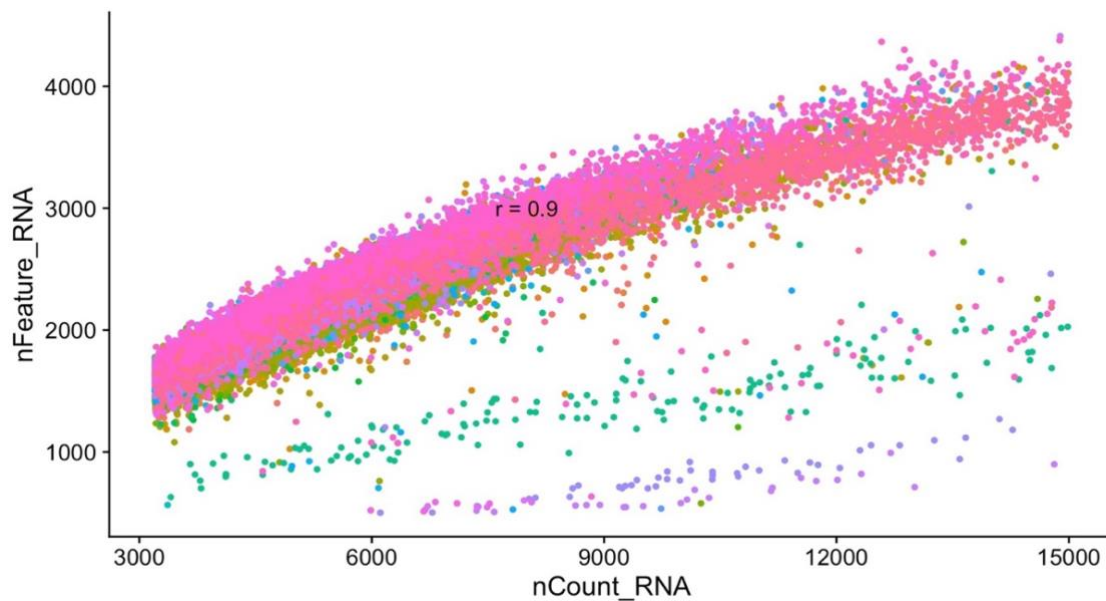


Figure 5: *UMI Count vs. Feature Count Scatter Plot*

The resultant plot is used visualize the correlation between the UMI Count (nCount_RNA) and number of unique genes (nFeature_RNA). The 'r' value represents Pearson's correlation coefficient.

3.2 Clustering & Annotation

To assess the variance explained by each principal component (PC) after PCA was run on the single cell dataset, an elbow plot was generated (Fig. 6). Initially, only 30 PCs were used, but the elbow was not fully established, so additional components were added. The midbrain cells identified after the first round of clustering were visualized on the UMAP structure, revealing a distinct group of cells in the clusters from early development at -5, 5 on the UMAP (Fig. 7). To ensure proper clustering of the cells that had cerebellar origin, these were removed from the final dataset. Several thresholds were tested for midbrain cell removal to find a balance between eliminating enough cells to avoid influencing downstream clustering results and retaining sufficient cells to preserve dataset integrity.

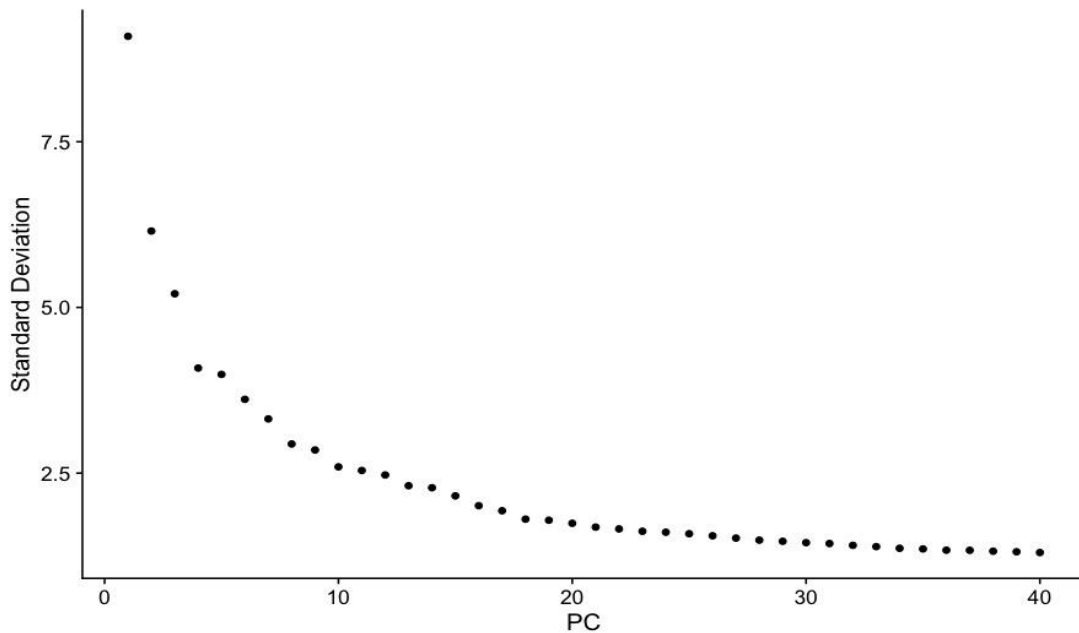


Figure 6: *Elbow Plot Following Principal Component Analysis on Final Seurat Object*
The resultant elbow plot shows how much of the total standard deviation observed in the dataset is attributed to each principal component. 40 principal components (PCs) were used, and the elbow, which represents the optimal number of PCs to retain, is observed near the 40th PC.

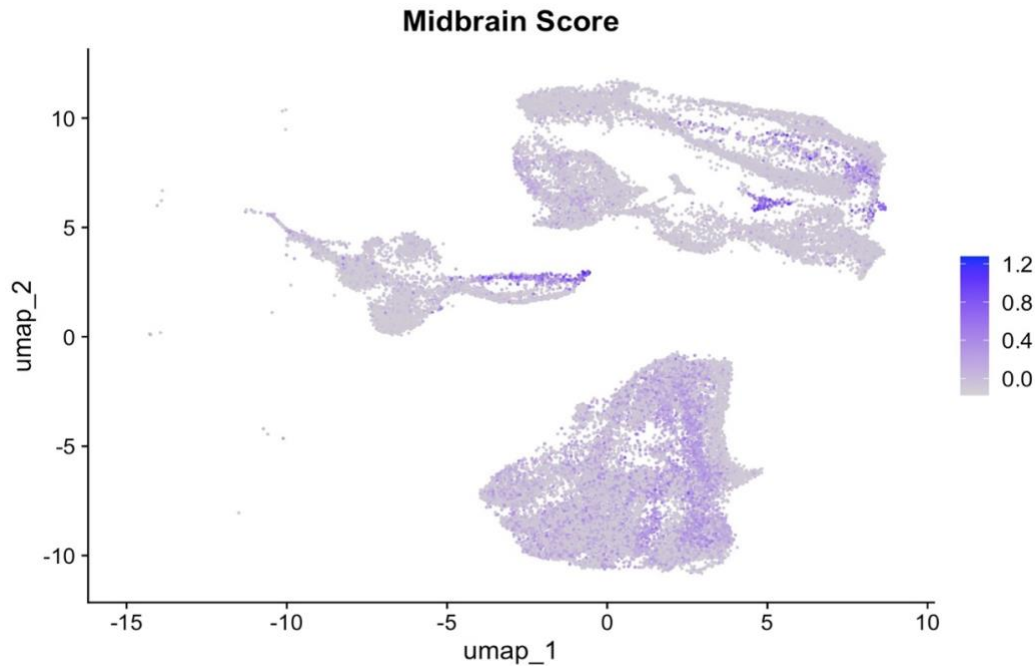


Figure 7: *Two-dimensional UMAP Visualization with Overlapping Midbrain Score*
The calculated midbrain score was visualized on the initial UMAP clustering structure to identify regions that may reflect groups of midbrain cells. Two main clusters were identified by dark-purple shading and were filtered out before the final iteration of clustering.

The resultant UMAP with 32 Seurat clusters and the annotated UMAP with corresponding cell types are seen in Figure 8 & 9. Clustering revealed three superclusters corresponding to distinct groups of cells: (1) early NPCs, (2) differentiating NPCs together with glial cells and neurons, and (3) GNPs/granule cells. The types of cells identified are aligned with the emergence of known cell types during cerebellar development. With the use of Loupe Browser, the co-expression of multiple features was visualized on the UMAP clusters and used to annotate cell types with a high degree of confidence. The grouping of similar cell types such as the GABAergic progenitors and neurons, the early NPCs, and the GNPs/Granule neurons, highlights the transcriptomic similarities between these cell types. To understand how the abundance changed as development progressed, a plot that showed the proportion of cell types at each developmental timepoint was generated (Fig. 10). This plot shows rapid emergence and

disappearance of cell types at different timepoints. Early progenitors made up a large proportion of cells in E10-E11, and they quickly differentiated to lineage specific progenitors, neurons and glial cells. After E15 and postnatally, GNPs/Granule neurons were the most abundant cell type.

The expression of master regulators *Atoh1* and *Ptfla*, expressed in RL and VZ progenitors, was plotted on the resultant UMAP to visualize expression patterns (Fig. 11). High *Atoh1* expression is seen in the early progenitor supercluster, where there is minimal expression of *Ptfla*. The expression of two additional genes associated with the GABAergic lineage, *Neurog2* and *Ascl1*, was also plotted on the UMAP. The expression of these two genes can be clearly seen in the early progenitor supercluster, and again in the GABAergic progenitor clusters.

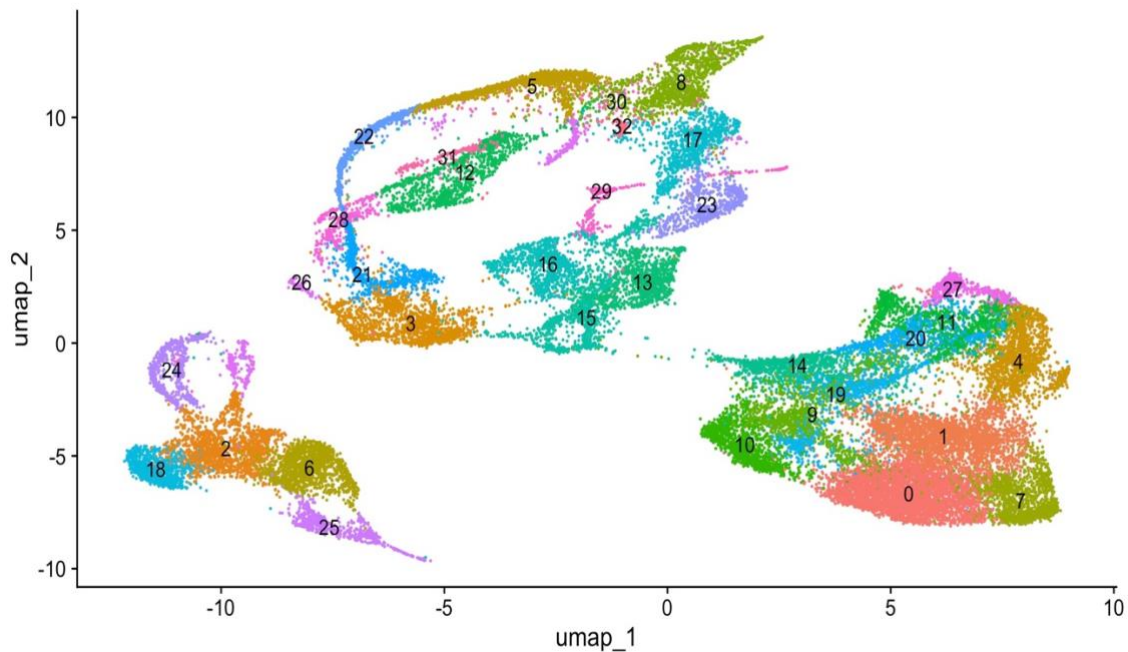


Figure 9: Final Two-dimensional UMAP Visualization with Seurat Clusters

Seurat UMAP clustering result using a resolution of 1.4. 32 clusters were identified by the clustering algorithm. Only clusters of cerebellar origin were kept.

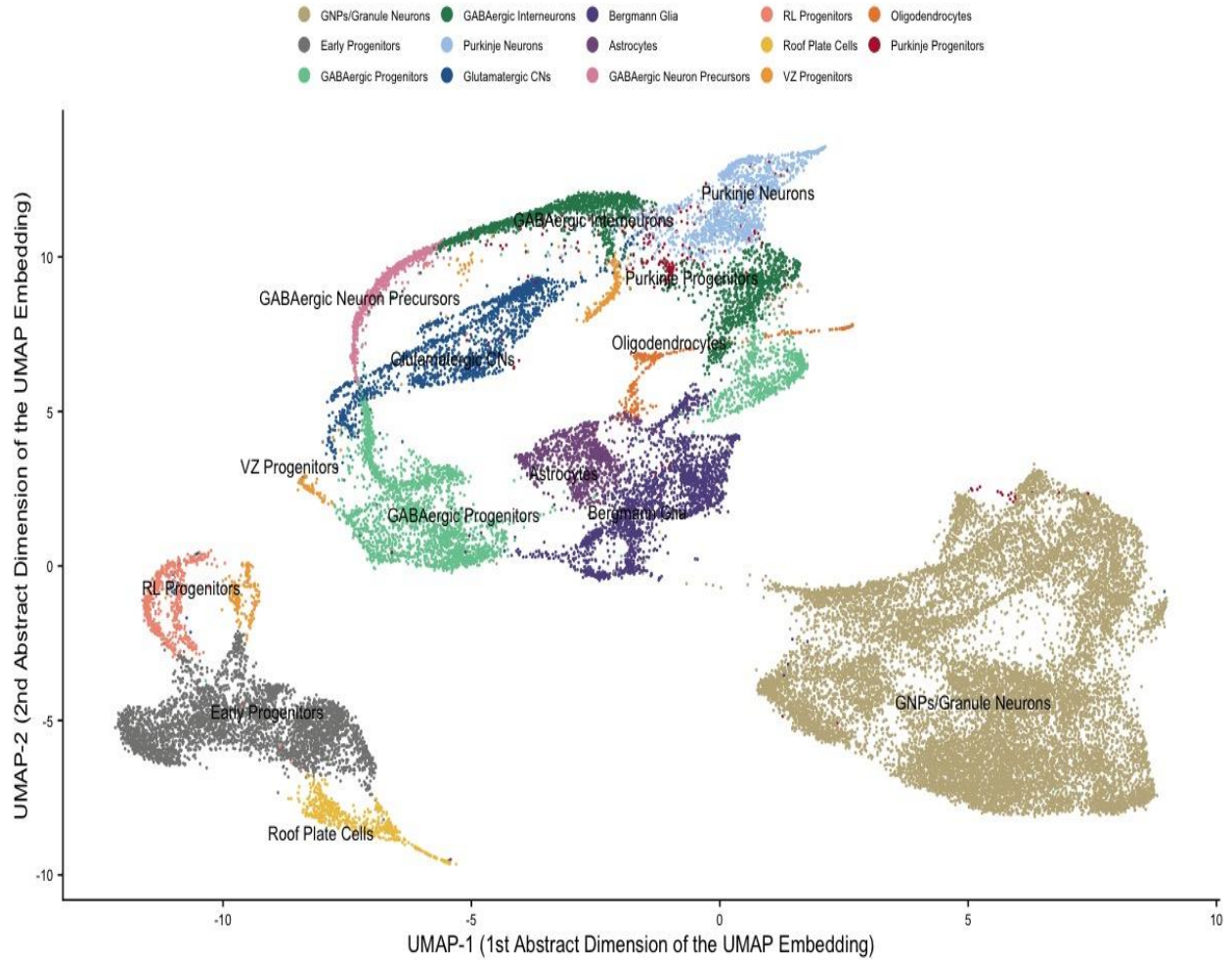


Figure 9: *Two-dimensional UMAP Visualization with Cell Type Annotations*

Seurat UMAP clustering result using a resolution of 1.4. The cluster identities were assigned using expression of known marker genes and estimated emergence of cell types during development. Only clusters of cerebellar origin were kept.

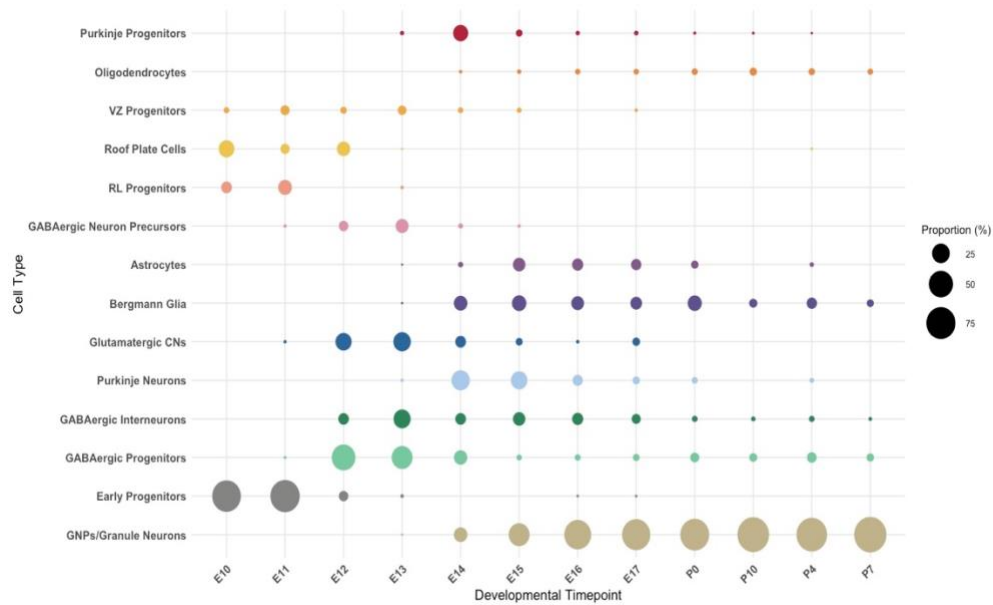


Figure 10: Proportion of Cell Types Across Developmental Timepoints

Relative proportions of inferred cell types at each of the sample timepoints (E10-P10). Cell type colours correspond to cell populations from Figure 9. Samples were collected less frequently postnatally.

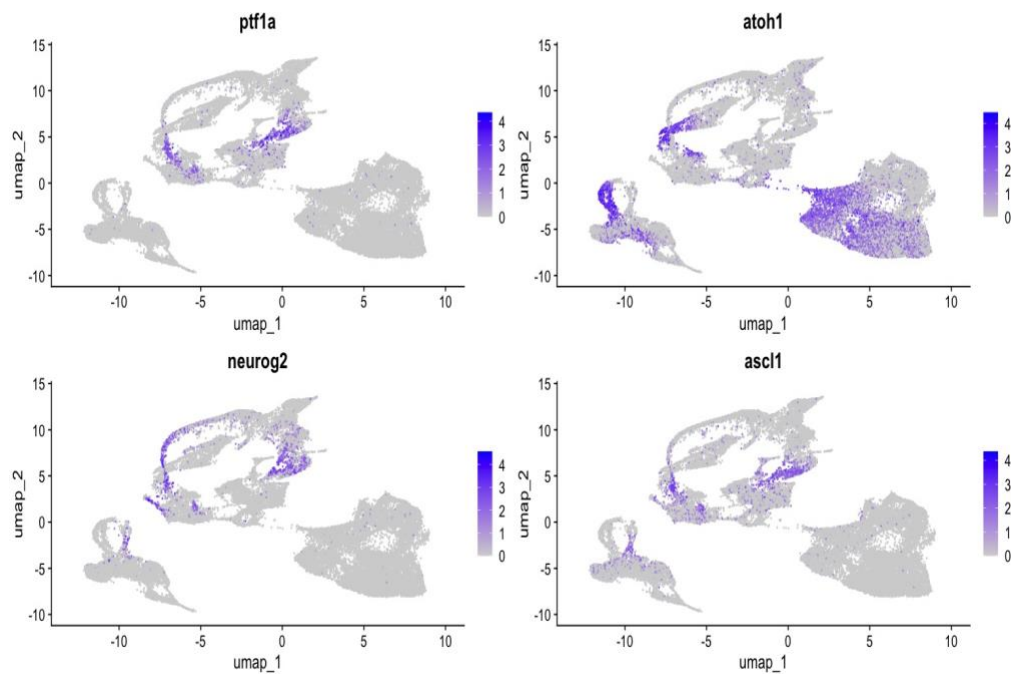


Figure 11: Expression of Key Early Progenitor Marker Genes

The relative expression of 4 marker genes visualized on the final two-dimensional UMAP graph to identify areas of upregulated gene expression. The legend is a scale of gene expression.

3.3 Early NPC Analysis

A heatmap of the top 12 marker genes for each of the distinct populations of early progenitors was generated, highlighting unique gene expression patterns for each cluster (Fig. 12). To add validity to this analysis, the difference in the proportion of cells that expressed a marker gene compared to all other cells was also used to select the top genes for visualization. Genes that are upregulated in a cluster but not unique to it are less informative than those that are both uniquely expressed and upregulated. The location of the early NPC populations can be seen in Figure 13. The boundaries were created based on the visualization of marker genes in Loupe Browser. Early NPC-1 emerged during E10, and the rest of the early progenitor populations emerged during E11. The full marker gene lists were used to perform GO analysis, and the top 5 enriched GO terms for each cluster were plotted (Fig. 14).

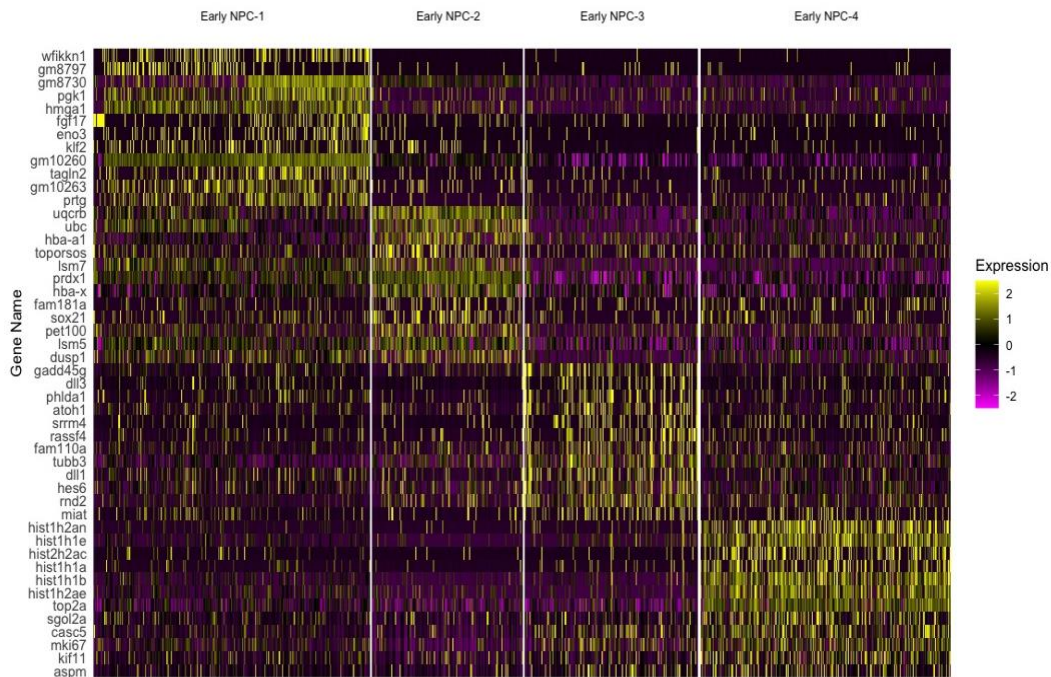


Figure 12: Heatmap of Top Marker Genes Between Early NPC Populations

Early NPC populations identified by re-clustering the original early progenitor cluster. Top 12 marker genes for each population of NPCs were selected based on avg_log2FC and pct.diff (cluster specific gene expression proportion – proportion in rest of the cells).

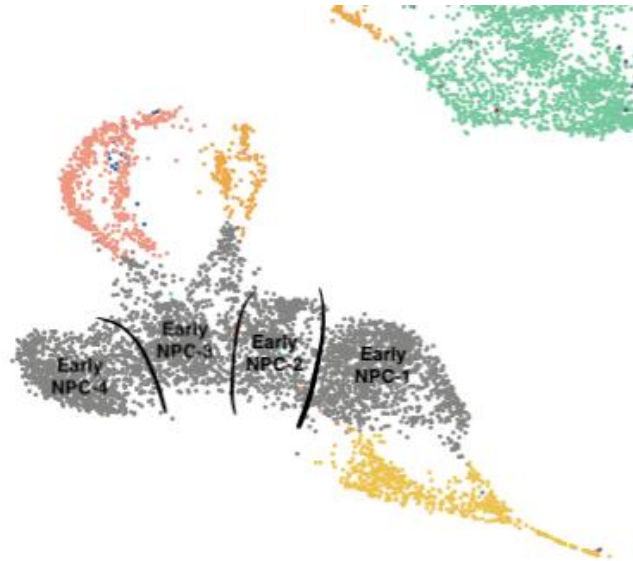


Figure 13: *Approximated Clusters of Early NPC Populations*
The original cluster of Early NPCs was re-clustered and the resultant distinct early NPC subclusters from Figure 10 are labelled on the original UMAP structure.

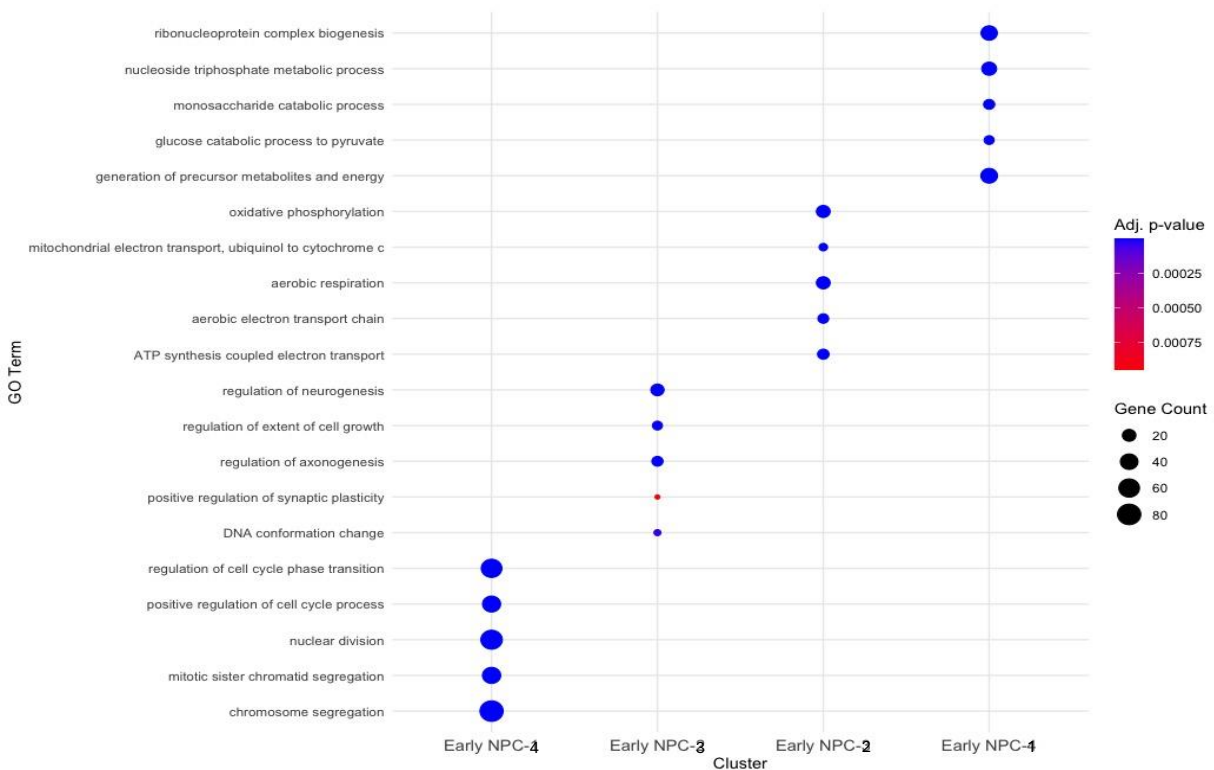


Figure 14. *Top GO Terms for Distinct Early NPC Populations*
Top 5 GO terms for each NPC cluster, highlighting distinct functional differences between populations. Top GO terms were scored based on a combination of number of genes per term and fold enrichment.

3.4 Pseudotime Analysis and Trajectory Inference

The final component of this project was the analysis of gene expression dynamics over pseudotime using Monocle3. After the principal graph trajectory was predicted from the dataset, it was plotted over the UMAP with the corresponding pseudotime progression (Fig. 15). The path shows that the ‘Early Progenitors’ are at the start of the trajectory, and they then differentiate into RL and VZ progenitors, and progress towards more mature neuron and progenitor populations. GNPs/Granule neurons arise later in pseudotime. The trajectory shows a branching point, where the trajectory follows two different paths, one towards a GABAergic lineage and a separate path towards a glial lineage (Fig. 16) The resultant trajectory was used to understand the specific dynamics occurring at this branching point. The top 5 genes showing the most significant expression differences across pseudotime between the glial population and GABAergic population of cells are shown in Figures 19 & 20.

Modules of co-expressed genes with similar expression dynamics over pseudotime were identified for the subset of cells representing the glial and GABAergic progenitor trajectories and for the supercluster of early progenitor cells (Fig. 17 & 18). The module visualizations highlight pseudotemporal regions with distinct genetic profiles, which can be interpreted as transitional cell states. The modules seen in Figure 18 strongly resemble the distinct populations of early NPCs that were identified previously. Each module’s gene list contains hundreds of genes and can be subjected to further analysis to understand the cells functional profile.

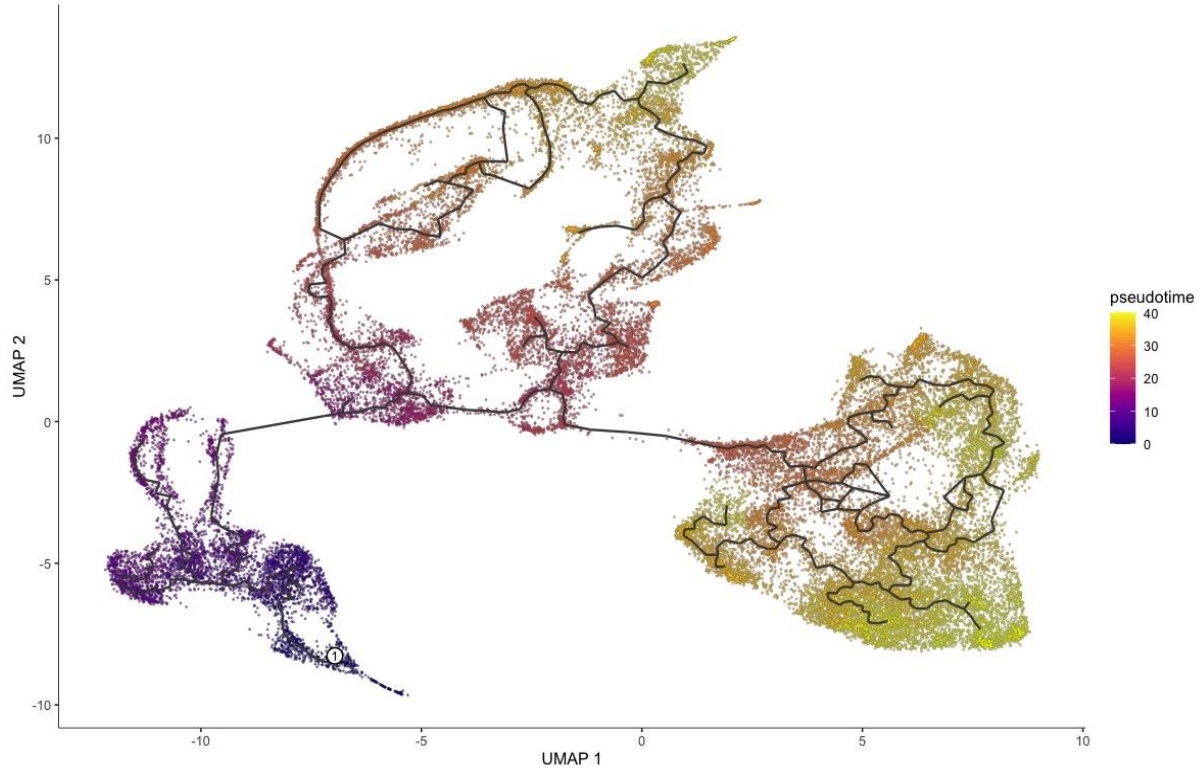


Figure 15. *Two-dimensional UMAP Visualization with Inferred Trajectory and Pseudotime*
To create one continuous trajectory, the clusters retained for trajectory inference and pseudotemporal inference were treated as one partition. Purple represents the start of pseudotime and orange/yellow represents cells that emerge later in pseudotime.



Figure 16. *Glial and GABAergic Clusters with Pseudotemporal Ordering*
Subset of the data containing Glial cells and GABAergic progenitors, showing the two distinct trajectories of cell fate. The trajectory shows a branching point towards a non-neuronal lineage.

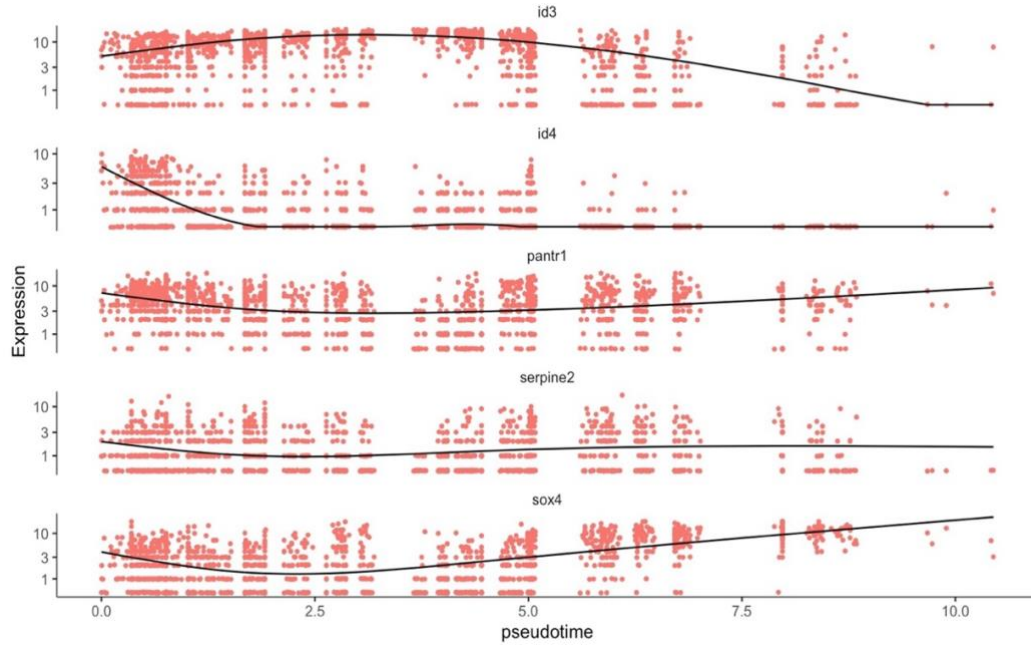


Figure 19. *Expression of Key Genes Involved in GABAergic Cell Fate Over Pseudotime*
GABAergic cell expression of genes that show significant expression dynamics over pseudotime between the GABAergic and glial trajectories from a common progenitor pool.

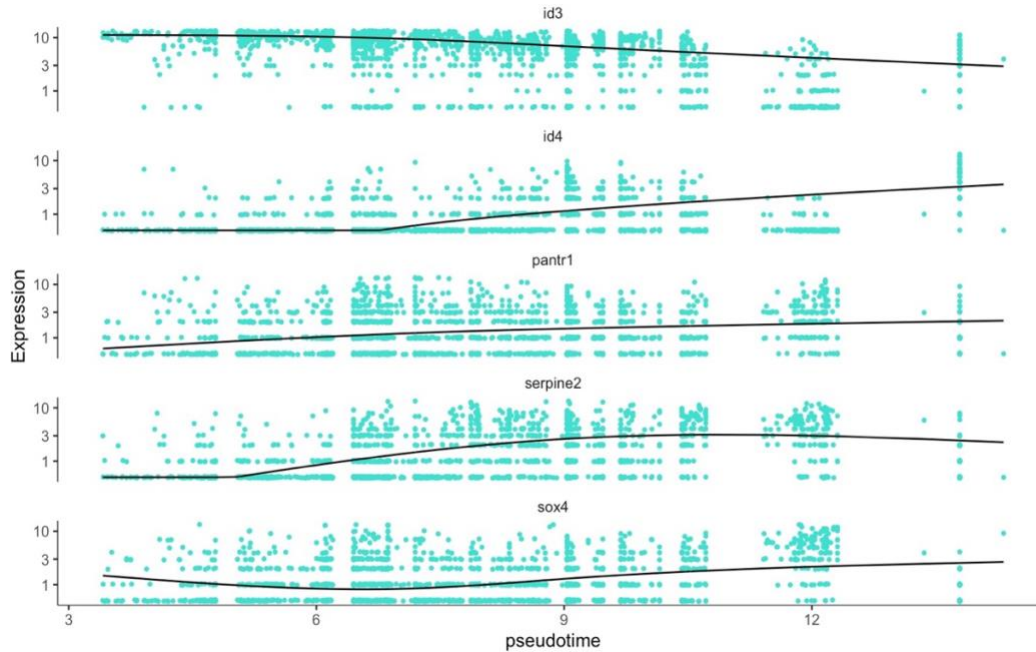


Figure 20. *Expression of Key Genes Involved in Glial Cell Fate Over Pseudotime*
Glial cell expression of genes that show significant expression dynamics over pseudotime between the GABAergic trajectory and glial trajectory from a common progenitor pool.

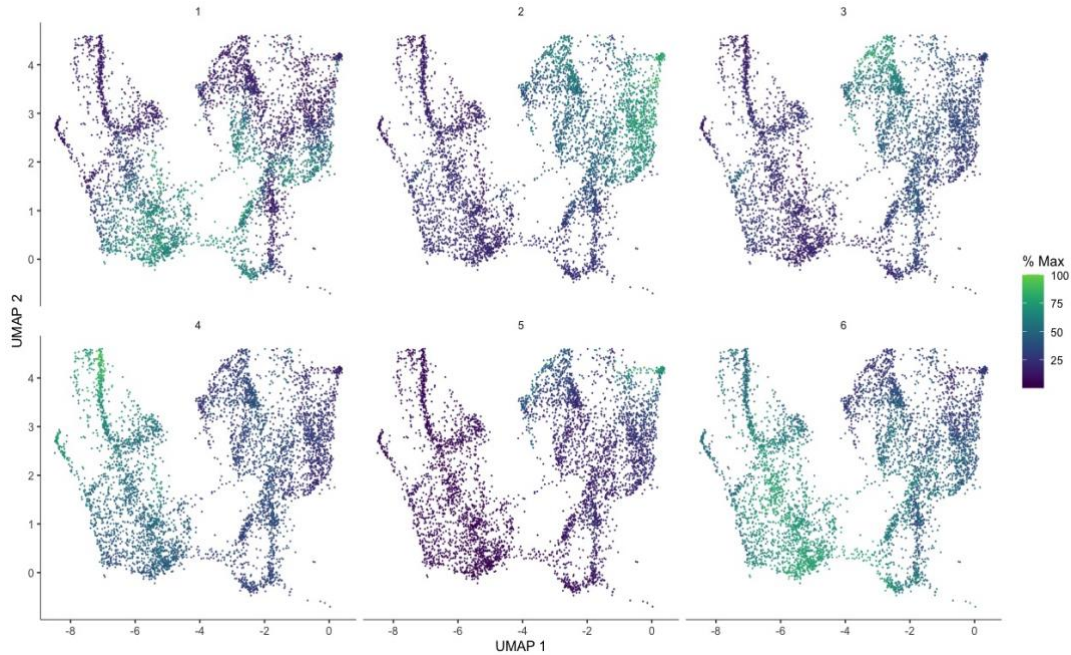


Figure 18. *Distinct Modules of Co-expressed Genes for GABAergic & Glial Lineages*
 Modules represent clusters of genes that share similar expression dynamics across pseudotime. Modules highlight fate dynamics. % Max represents relative expression for each module.

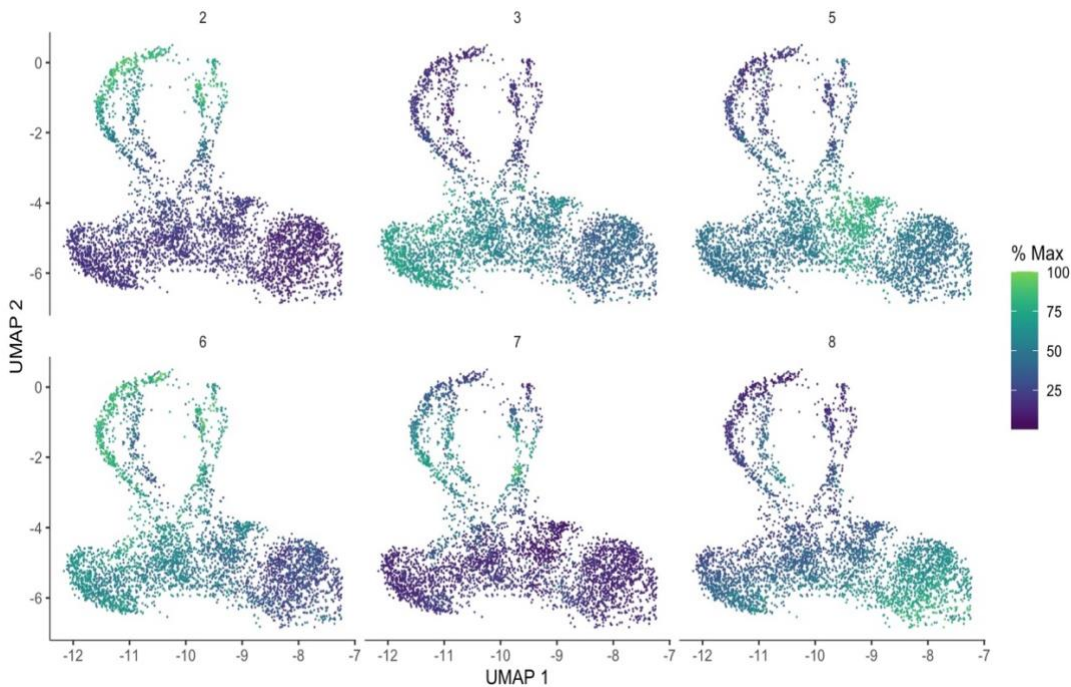


Figure 17. *Distinct Modules of Co-expressed Genes for Early NPCs*
 Modules represent clusters of genes that share similar expression dynamics across pseudotime. Modules highlight early fate dynamics. % Max represents relative expression for each module.

4.0 Discussion

After completing all quality filtering steps, the final cell count of 36,230 was closely aligned with the 39,245 cells reported by Carter et al. (2018). A key difference between the published findings and this analysis was the lower cell counts for the E12 samples. This was a feature of the raw count data and attempts to download the raw reads were not successful as the authors did not upload both paired-end read files. The remaining samples had high cell counts, which allowed for a robust analysis. The emergence of known cerebellar cell types seen in Figure 10 aligns with literature and with the results produced by Carter et al. (2018).

Atoh1 expression in cells was observed from E10-E12 and again postnatally, which is reflective of previous findings. The expression of *Ptf1a* starting at E12- E13 also corresponds to previous literature. High expression of *Ascl1* and *Neurog2* in the cluster emerging at E10-E11 labelled ‘VZ progenitors’, and later in the GABAergic progenitor populations was unexpected. *Ptf1a* expression is essential for the development of the GABAergic lineage, so the expression of *Ascl1* and *Neurog2* prior to *Ptf1a*, suggests that these genes are involved in establishing this lineage (Hoshino et al., 2005). Understanding how these genes are involved in the development of the GABAergic lineage would be a valuable next step to expand on these findings.

The novel components of this analysis include the characterization of distinct early NPC populations and identification of key genes implicated in the bifurcation of NPCs toward GABAergic progenitor and glial identities. Modules of co-expressed genes showing transitional cell states were isolated for both the supercluster of early NPCs and the subset of cells where the trajectory branches to GABAergic progenitors and glial cells. Although further analysis is required to understand the transcriptional dynamics for each module, the modules identified in Figure 17 closely resemble the 4 early NPC populations identified in Figures 12/13.

The top marker genes identified for Early NPC-1, as seen in Figure 12, include *Fg17*, *Hmg1*, *Klf2*, *Pgk1* and *Eno*. *Fg17* serves as an indicator of active signaling and patterning and may reflect cellular interactions with the surrounding environment. *Hmg1* and *Klf2* are transcription factors that regulate gene expression and *Pgk1* and *Eno3* are glycolysis related genes (National Center for Biotechnology Information, 2025). Based on the top GO terms: ‘glucose catabolic process to pyruvate’, ‘nucleotide triphosphate metabolic process’ and ‘ribonucleoprotein complex biogenesis’, this population is showing high metabolic activity, energy production (ATP) and active protein translation. Early NPC-1 was the first progenitor cluster to emerge and likely represents a group of cells undergoing rapid growth and increased activity as they prepare to proliferate and/or differentiate. Previous work has shown that the process of neural differentiation involves metabolic shifts and epigenetic changes that allows the cell to generate energy in anticipation of fate decisions (Sánchez-Ramírez et al., 2024).

With respect to ‘Early NPC-2’, *Prdx1* and *Dusp1* are genes involved in a cell’s response to oxidative stress and cellular signalling pathways (National Center for Biotechnology Information, n.d.). *Sox21* was also identified as a marker gene, which has been linked to neural differentiation. *Sox21* is believed to act contextually and may be regulated by environmental factors in different parts of the nervous system. A gain-of-function experiment showed that *Sox21* is involved in progenitor maintenance and in the rate of differentiation. The absence of *Sox21* is linked to cell death, due to a cells inability to differentiate after exiting the cell cycle (Whittington et al., 2015). The corresponding GO terms: ‘aerobic respiration’, ‘oxidative phosphorylation’ and ‘ATP synthesis coupled electron transport’ are indicators of a population of cells with high activity in the mitochondrial oxidative phosphorylation pathway. This population is generating ATP through aerobic respiration and may be seen as maintaining efficient energy

production. In combination with the presence of *Sox21*, this population may be using lots of energy to maintain its progenitor identity and self-renewing abilities before establishing a defined fate. These findings further implicate *Sox21* as a key regulatory gene involved in maintaining a cells capacity to proliferate. The importance of metabolic changes in relation to neural lineage commitment has been emphasized in previous work. The differentiation of NPCs to neurons involves metabolic reprogramming, where the main source of energy switches from glycolysis to oxidative phosphorylation (Sánchez-Ramírez et al., 2024).

The 'Early NPC-3' population shows clear signs of differentiation, based on the marker genes *Atoh1* and *Barhl1*, which are transcription factors involved in neuronal fate decisions. *Atoh1* is recognized as an indicator of a glutamatergic fate and is expressed by rhombic lip progenitors (Machold et al., 2011). *Dll3* is linked to notch signalling and the coordination of differentiation and proliferation, highlighting a progressive differentiation of cells towards a neuronal identity (National Center for Biotechnology Information, n.d.). Associated GO terms: 'regulation of neurogenesis', 'regulation of axonogenesis' and 'regulation of extent of cell growth' are implicated in the formation of new neurons and axon growth. It is well established that transcription factors which regulate specific transcriptional programs are needed for neuronal differentiation (Sánchez-Ramírez et al., 2014; Carter et al., 2018). This cluster is located beside the clusters labelled 'RL Progenitors' and 'VZ Progenitors', and the pseudotime trajectory shows the progression of these cells towards lineage specific progenitors.

In the 'Early NPC-4' population, the key markers are mostly histone genes, including *Hist1h2an*, *Hist1h1e*, *Hist1h1a*. *Ckap5* is gene that codes for a protein essential for mitotic spindle formation and microtubule dynamics (National Center for Biotechnology Information, n.d.). The top marker genes for this cluster are related to cell cycle and mitosis, and based on

this, these cells are likely in proliferative phase. These cells are actively growing and dividing to create more cells. They are likely in a differentiated state and are now proliferating, or they are still unspecialized and proliferating before establishing a defined fate. The top GO terms include: ‘nuclear division’, ‘DNA confirmation change’, ‘mitotic sister chromatid segregation’, and ‘regulation of cell cycle progress’, which all support a proliferating identity.

These results are aligned with the location of the modules of co-expressed genes identified in this supercluster of early NPCs (Fig. 17). These findings show that there are distinct functional differences between these populations of early NPCs. Each cluster represents a key stage in the progression and differentiation of early NPCs during cerebellar development. This process may be highly complex, and groups of cells representing different transitional states may emerge simultaneously. Based on the trajectory produced by Monocle, there may not be just one path, and cells may remain plastic for stochastic fate switching before fully committing.

The 5 genes that showed the most significant expression differences as the GABAergic progenitor lineage and glial lineage diverged were *Id3*, *Id4*, *Pantr1*, *Serpine2*, *Sox4*. *Serpine2* is a serine protease inhibitor, which binds to serine proteases and reduces their proteolytic potential (National Center for Biotechnology Information, n.d.). This molecule is secreted primarily by astrocytes and glial cells in the CNS and is a modulator of neurite outgrowth, differentiation, extracellular matrix remodeling and is linked to glial maturation (Pagliara et al., 2014). The expression trends relative to the GABAergic lineage and glial lineage show increased expression in glial cells. A strong positive linear relationship is observed as the glial lineage progresses through pseudotime, whereas an overall steady expression profile is observed in the GABAergic progenitor trajectory.

Sox4 has been shown to play a role in progenitor cell differentiation and neuronal maturation. *Sox4* is an important molecular switch that drives neural development and is one of the Group-C *Sox* genes, which through a cascading effect, direct progression towards a neuronal phenotype (Bergsland et al., 2006). The expression of *Sox4* is showing a positive correlation over pseudotime for both the GABAergic and glial lineages as both cell types progress to more differentiated states. These results highlight the importance of *Sox4* in neuronal development for both GABAergic and non-neuronal cells.

A recent study has linked *Pantr1*, a long intergenic noncoding RNA (lncRNA), to *FOXG1*-syndrome. *FOXG1*-syndrome is a rare autism-like disorder, and *FOXG1*'s role is to maintain progenitor levels by preventing premature differentiation in both human and mice (Gather et al., 2025). Cells with a *FOXG1* deficiency showed impaired development and increases post mitotic cell death. *FOXG1* was found to be an upstream regulator of *Pantr1* expression and together, are involved in the formation and development of dendritic structures in neurons (Gather et al., 2025). The regulation of these genes is important for controlled differentiation, and impacts progenitor progression to a neuronal identity. The expression of this gene in both the GABAergic and glial lineage is increasing slowly over pseudotime.

Id4 has been shown to play a role as a key regulator in neural stem cell proliferation and differentiation in the developing cortex and hippocampus. *Id4* inhibits the proliferation of neurons that are differentiating through the *RBI* pathway and in *Id4* deficient mice, the ventricular zone of the neocortex and early hippocampus showed a 20-30% reduction in mitotic neural precursor cells (Bedford et al., 2005). No current research has discussed the role of *Id4* in the differentiation of NPCs to glial cells in the cerebellum, but the increase in *Id4* expression as the glial lineage matures, and decrease as the GABAergic lineage progresses, is noteworthy and

potentially highlights a mechanism required for glial differentiation. The other member of the *Id* family, *Id3*, has a role in the maintenance of progenitor population levels by inhibiting differentiation and regulating this transition (Gupta et al., 2020). Progenitors that are undergoing proliferation eventually need to differentiate, and the coordination of this progression is crucial. A decreasing expression trend is observed for both the GABAergic and glial lineages over pseudotime, and a decrease in *Id3* expression is important for neuronal development.

To summarize, four early NPC populations with distinct functional profiles were identified. Early NPC dynamics were uncovered by tracking changes in gene expression over pseudotime during a branching event between GABAergic progenitors and non-neuronal glial cells. Finally, transitional cell states with similar co-expression patterns were identified and can be further analyzed to characterize key developmental stages in cerebellar development.

Future steps to build on this analysis would include doing GO analysis on the modules of co-expressed to understand the functional states of these transitional populations. This would provide a detailed view of the transcriptional mechanisms at play at specific stages in cerebellar development. To further understand how the early NPC populations start differentiating, a similar branching analysis can be done to identify gene expression dynamics between a VZ and RL progenitor fate. Knockout mouse models can be used to assess whether specific lineages mature normally in the absence of candidate regulators such as *Id4* and *Serpine2*. Additionally, over- and under-expression of key regulators can be tested to understand how they influence the progression of lineage maturation. Lastly, variants found in the genomes of individuals with neurodevelopmental disorders could be scored based on their location and frequency in respect to the genes and modules identified in this analysis. This information could then be used to develop genetic testing panels (I. Tobias, personal communication, July 29, 2025).

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